Charge trapping in SiO2-MoS2 TENGs

"How does pore size affect charge trapping in SiO2-MoS2 TENGs?" *Generated by LeapSpace*

Pore size generally affects charge trapping in SiO_2–MoS_2 TENGs nonlinearly: introducing pores can create more trapping sites and raise trapped charge density, but pores that are too large or too open can weaken charge retention and leakage resistance. In this system, the most supportable target is **moderate, well-distributed mesoporosity**, not maximum pore size. [11] [101] [102] [103]

How pore size changes trapping behavior

Pore-size regime	Expected effect on charge trapping	Why
Very small nanopores / dense mesopores	More trap sites, but limited accessibility if too constrained	Nanopores can act as charge-trapping centers, with charges aggregating along pore boundaries and staying trapped; in silica systems, very small pores can also be less effectively utilized if transport into them is difficult. [101] [102] [104]
Moderate mesopores	Best balance of trapping and retention	Moderate porosity increases interfacial area and trap density while preserving enough structural continuity for stable charge storage. This matches TENG evidence that charge density and trapping improve with porous structures, and prior conversation evidence that intermediate porosity is preferable. [46] [48] [105]
Large / highly open pores	Trapping can weaken in practice	Larger pores can reduce effective control of charge and create easier pathways for loss; in porous dielectrics, excess porosity lowers dielectric stability and can worsen retention. [106] [107] [108]

What this means for SiO_2–MoS_2

- **The MoS_2/SiO_2 interface is itself a strong trapping region**, with high-density interface states and long charge retention already reported for MoS_2–SiO_2 systems. [59] [103] [109]
- **Pore size matters mainly because it changes how much interfacial area is available** and how easily charges can be stored at pore boundaries and interfaces. Smaller pores increase boundary area, but only up to the point where transport, continuity, and retention remain favorable. [48] [101] [107]

- For a **SiO_2–MoS_2 TENG**, the most defensible design is a **thin, continuous MoS_2 shell over a moderately mesoporous SiO_2 core or shell**, because that supports trapping without sacrificing the barrier function that helps suppress charge decay. [11] [103] [110]

Bottom line

Smaller pores help create more trapping boundaries, but the best charge trapping in SiO_2–MoS_2 TENGs is likely achieved with moderate mesopore sizes that maximize interfacial area while preserving shell continuity and charge retention. The evidence does **not** support simply making pores as small or as large as possible. [11] [101] [102] [103] [105]

Confidence level: **Medium** — the evidence directly supports pore-boundary trapping in porous dielectrics and strong MoS_2/SiO_2 interfacial trapping, but it does not include a direct pore-size sweep for one exact SiO_2–MoS_2 TENG formulation.

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